

Solution
HUMAN EYE -03
Class 10 - Science
Section A

1. State True or False:

- (i) **(a)** True
Explanation: True
- (ii) **(a)** True
Explanation: True
- (iii) **(a)** True
Explanation: True
- (iv) **(a)** True
Explanation: True
- (v) **(a)** True
Explanation: True
- (vi) **(a)** True
Explanation: True
- (vii) **(a)** True
Explanation: True
- (viii) **(a)** True
Explanation: True
- (ix) **(b)** False
Explanation: False. If the size of the scattering particles is large enough, then, the scattered light may even appear white.
- (x) **(b)** False
Explanation: False. Iris is present behind the cornea.
- (xi) **(a)** True
Explanation: True
- (xii) **(b)** False
Explanation: False. The lens of human eye forms an image on a light-sensitive screen called the retina.
- (xiii) **(a)** True
Explanation: True
- (xiv) **(a)** True
Explanation: True
- (xv) **(b)** False
Explanation: False. Cataract is a condition in which the crystalline lens of people at old age becomes milky and cloudy resulting in blurred vision.
- (xvi) **(a)** True
Explanation: True
- (xvii) **(a)** True
Explanation: True
- (xviii) **(b)** False
Explanation: False. The splitting up of white light into seven colours on passing through a transparent medium like a glass prism is called dispersion of light.
- (xix) **(a)** True
Explanation: True

Section B

2. Fill in the blanks:

- (i) 1. Scattering
- (ii) 1. Atmospheric refraction
- (iii) 1. Seven
- (iv) 1. Accommodation
- (v) 1. Myopia
- (vi) 1. Tyndall effect
- (vii) 1. Deviation
- (viii) 1. Presbyopia
- (ix) 1. Dispersion
- (x) 1. Iris
- (xi) 1. Cataract
- (xii) 1. Rainbow
- (xiii) 1. Spectrum
- (xiv) 1. Retina
- (xv) 1. Refraction
- (xvi) 1. Pupil
- (xvii) 1. 2 minutes
- (xviii) 1. Isaac Newton
- (xix) 1. Concave

Section C

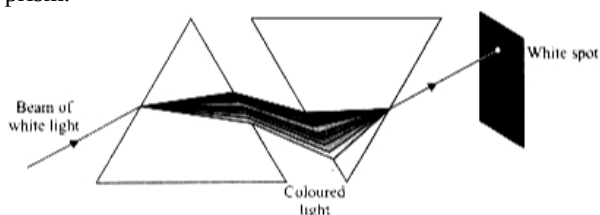
3. i. The 3 common refractive defects of vision are

- a. Myopia (or nearsightedness) It is corrected by using concave lenses of appropriate power which diverge the refracted rays and thus the image will be formed onto the retina instead of in front of retina.
- b. Hypermetropia (or farsightedness) It is corrected by using convex lenses of appropriate power which act as converging lens and the image is received onto retina not beyond it.
- c. Presbyopia which is caused by ciliary muscle defect and corrected by using bifocal lenses of appropriate power in which the upper part consists of a concave lens (to correct myopia) and lower part consists of convex lens (to correct hypermetropia).

ii. Eyes of a dead person can be donated to the person having corneal blindness, it will help him/her to see the world. We can register ourselves and also encourage others also to register to eye donation camps who can preserve our eyes after our death and donate them to the needy.

4. In eye, the distance between lens and the retina remains the same, while crystalline lens automatically changes its focal length (by changing its curvature due to pull of push of ciliary muscles) according to the distance of the object so as to bring the image to a sharp focus upon the retina. While seeing the far off objects such as distant tree, the eye lens becomes thinner and flatter so as to increase its focal length. To see the object close to the eyes such as printed page, the lens becomes thicker so as to decrease its focal length. The process by which the eye can adapt itself to see objects at different distances is called accommodation.

5. The phenomenon of splitting of white light into its constituent colours on passing through a prism is known as the dispersion of white light. This splitting of the light rays occurs because of the different angles of bending for each colour and these different angles of bending occur because different component of light faces different refractive indices when passing through the glass prism.



It is essential that the two prisms used for the purpose should be identical and placed in an inverted position with respect to each other so that the second prism completely nullifies the dispersion caused by the first prism and we get pure white light.

6. Here, $u = -25$ cm (Normal near point)

$v = -1$ m = -100 cm (Near point of defective eye)

$f = ?$

Using lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or } \frac{1}{-100} - \frac{1}{-25} = \frac{1}{f}$$

$$\text{or } -\frac{1}{100} + \frac{1}{25} = \frac{1}{f}$$

$$\text{or } \frac{-1+4}{100} = \frac{1}{f}$$

$$\text{or } f = \frac{100}{3} = 33.3 \text{ cm}$$

Thus, the convex lens of focal length +33.3 cm is required to correct defect.

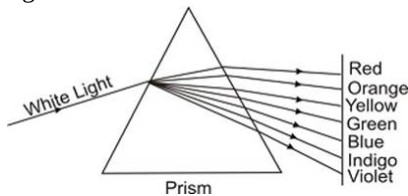
Here, $f = 33.3$ cm or 0.33 m

Now, Power $P = \frac{1}{f(\text{in m})}$

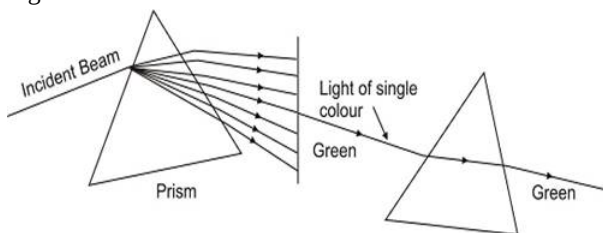
$$\frac{1}{+0.33} = +3.0 \text{ D}$$

Thus, the convex lens of power +3.0 D is required.

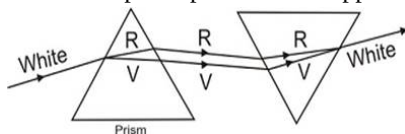
7. Dispersion of light: If a beam of white light falls upon one of the faces of the prism, it is seen in the other face of the prism that light has split up into seven colours. These colours are red, orange, yellow, green, blue, indigo and violet. The splitting of white light into its constituent colours is called dispersion and the coloured band obtained upon the screen is called a spectrum.



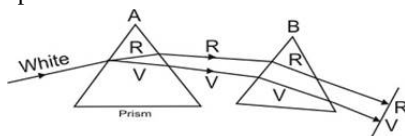
To show that colours are not produced by the prism but are present in white light itself and the prism only separates these, Newton isolated (separated) a particular colour say green. He placed another prism in the path of green beam. No further splitting of colour took place. The light was only deviated more. It clearly showed the prism just separates, a large number of colours coming together as white.



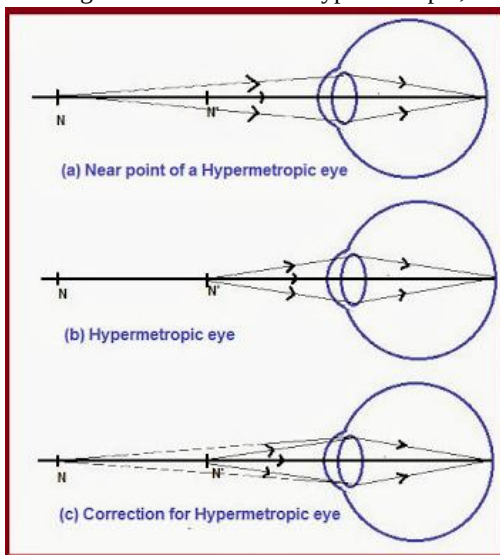
If a second prism exactly of same angle with its refracting edge opposite to the first is placed as shown, it is found that it results in white light again. The second prism deviates the rays in opposite direction. Thus dispersion produced by one prism is cancelled by the second prism placed in the opposite direction.



If the second prism is placed in the same way as the first, the coloured rays are again obtained on the screen but these are more spread.



8. For diagram of correction of hypermetropia,



Numerical

$$u = -25 \text{ cm}, v = -1 \text{ m} = -100 \text{ cm}$$

$$\frac{1}{7} = \frac{1}{v} - \frac{1}{u} = \frac{1}{-100} - \frac{1}{(-25)} = -\frac{1}{100} + \frac{1}{25} = \frac{-1+4}{100} = \frac{3}{100}$$

$$f = \frac{100}{3} \text{ cm} = \frac{1}{3} \text{ m}$$

$$P = \frac{1}{f} = \frac{1}{\frac{1}{3}} = 3 \text{ D}$$

Power = +3D convex / converging lens of +3 D.

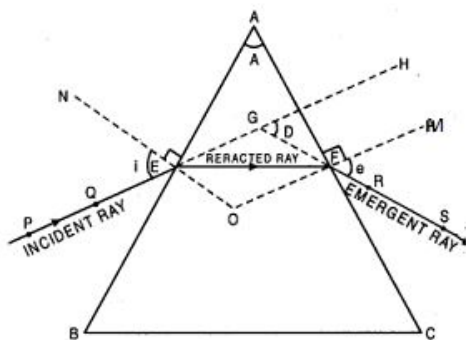
9. In a glass slab, the emergent ray is parallel to incident ray although emergent ray is displaced laterally. The lateral displacement depends upon the thickness of slab.

A glass prism has two triangular bases and three rectangular lateral surfaces.

Fix a sheet of paper on drawing board. Place a prism ABC on paper and draw its boundary. Draw a line PE and fix two pins P and Q on it. See from face AC of the prism and fix two more pins R and S so that P, Q, R, S all appear to be in same straight line.

Remove pins and the glass prism. Extend PE towards EH. Join R, S and produce back to meet EH at G. Draw OEN perpendicular to AB and MFO perpendicular to AC. Then PE is incident ray, EF refracted ray and FS, the emergent ray. $\angle PEN = \angle i$ is the angle of incidence, $\angle MFS = \angle e$, the angle of emergence and $\angle HGF = \angle D$, the angle of deviation. If $\angle BAC = \angle A$, the angle of prism, it can be proved that :

$$\angle i + \angle e = \angle A + \angle D$$



10. Myopia: The defect of an eye in which it cannot see the distant objects clearly is called myopia. A person with myopia can see nearby objects clearly.

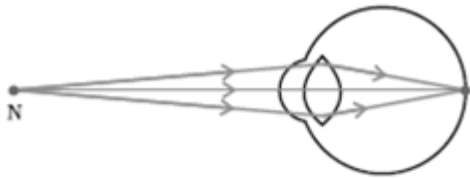
Hypermetropia: Hypermetropia is also known as long-sightedness. In this defect, a person can see the distant objects clearly but cannot see the nearby objects clearly.

Causes of Hypermetropia:

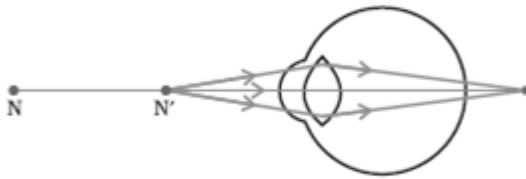
- i. The focal length of the eye lens is too long.
- ii. The eyeball has become too small.

In hypermetropia, the image of a distant object is formed behind the retina and not on the retina. The defect is corrected by using the Convex lens of suitable power so that the lens will bring the image back on to the retina. The ray diagram of hypermetropia

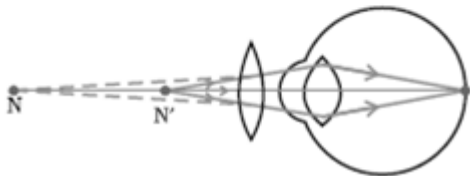
eye are as follows:



(a) Near point of a Hypermetropic eye

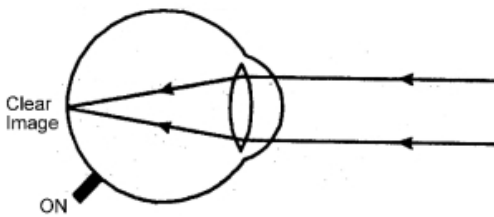


(b) Hypermetropic eye

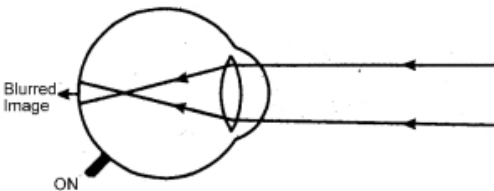


(c) Correction for Hypermetropic eye

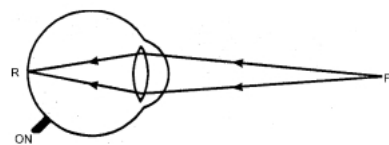
11. i. Scattering occurs when a light ray passes through an imperfect medium (medium which have particles in it, which act as scatterer) and get deflected from its straight path and scatters in many directions.
The colour of scattered light depends on the size of scattering particles and wavelength of light.
i.e. Scattering $\propto d^6$
[where, d = diameter of particle]
and scattering $\propto \frac{1}{\lambda^4}$
[where, λ = wavelength of particle]
- ii. The size of particles in the atmosphere is smaller than the wavelength of visible light, so they are more effective in scattering the light of shorter wavelengths, i.e. blue light. Thus, sky appears blue during the day.
- iii. For an astronaut, sky appears dark because there is no scattering of light in space due to absence of particles.
12. The person is suffering from myopia or short-sightedness.
Myopia or short-sightedness: In normal eye, the far point is at infinity. The rays coming from distant object (at infinity) gets focused on retina. In turn, the retina conveys the message to brain via optic nerve.



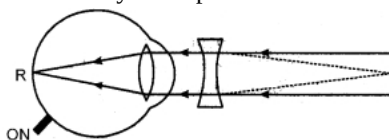
Normal eye: Far point at infinity. Rays from distant object meet at retina.



Defective eye: Eye ball is enlarged, or focal length decreased, cannot focus rays from infinity at focus.



Defective eye: Far point is nearer than infinity when at far point F.



Corrected eye: Concave (or any divergent) lens diverges the parallel rays from infinity to an extent that they appear to diverge from F. They get focused at retina.

Causes of defect in defective eye: The defect arise due to either :

- i. the length of eyeball of eye lens has decreased.
- ii. the focal length of eye lens has decreased.

Due to either or both the causes, the eye is not able to focus the rays from distant object at retina. Focusing is there at a point O in front of retina. Image formed on retina is blurred.

The defective eye is however able to focus the object upto its far point.

Correct of defect: We have observed that the defective eye has not been able to focus less inclined rays but has been able to focus more inclined rays.

To correct this defect, a concave or a divergent lens of appropriate focal length is placed in front of the eye so that parallel rays coming from infinity appear, after refraction through the lens, to come from the far point when eye is unaccommodated. These parallel rays from a distant object shall be brought to focus upon the retina. Obviously, the focal length of such a lens is equal to the distance of the far point from the eye.

13.	Myopia	Hypermetropia
	In this defect, a person can see the objects lying at short distances clearly but cannot see the far objects distinctly.	In this defect, a person can see the objects lying at long distances clearly but cannot see the nearby objects distinctly.
	Myopia is also called near\shortsightedness.	Hypermetropia is also called far\longsightedness.
	Myopia may arise due to a decrease in the focal length of eye lens.	Hypermetropia may arise due to an increase in the focal length of the eye lens.
	The size of eye ball becomes too long.	The size of eye ball becomes too short.
	Myopia is corrected by using a concave lens of suitable power.	Hypermetropia is corrected by using a convex lens of suitable power.

14. a. Function of human eye:

- i. **Cornea:** It focuses light rays entering into the eyes.
- ii. **Iris:** It controls amount of light entering the eyes and controls the size of pupil.
- iii. **Crystalline Lens:** It focuses the light rays entering into the eye and forms the image on retina.
- iv. **Cilliary Muscles:** It changes the shape of the lens for focusing.

b. In early morning, sun light has to travel the larger distance through the atmosphere to reach us. So, the shorter wavelengths scatter out. Only the longer wavelengths like red reaching us due to which sun appears red.

No, this phenomenon will not be observed by an astronaut on the Moon.

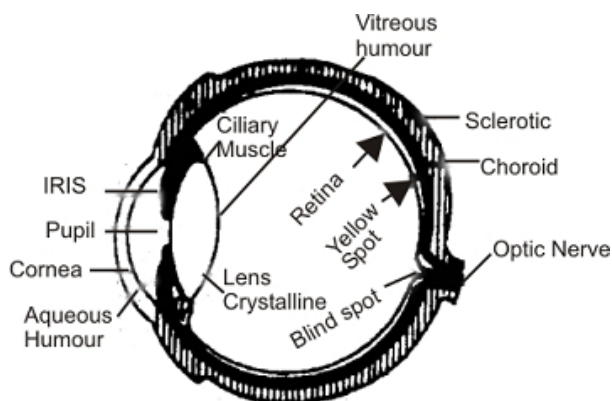
Cause: Moon has no atmosphere.

15. Sensitive portion of retina has large number of cells; one rod shaped and other cones shaped. Rod shaped cells are sensitive to the intensity or brightness of the light whereas cone shaped cells are sensitive to colours. Different animals have different type of cells. Bee has certain cone shaped cells which are sensitive to light beyond violet. This part of spectrum beyond violet called ultra violet region can be seen by bee whereas human being cannot see ultra-violet region. Chickens have very few rod shaped cells and hence their eyes are not able to see less bright light. Hence they stroll only in bright light. They come late in the morning and go back earlier in the evening. Most of its cells are cone shaped and hence are very sensitive to colours.

16. Human eye is the most remarkable and most delicate optical instrument.

Though in principle, it is similar to a photographic camera, it is far most delicate and perfect than the finest camera ever designed by human ingenuity.

Structure : The human eye consists of nearly spherical ball of about 2.5 cm diameter. Its outermost coating is made of a tough and opaque white substance known as SCLEROTIC. Sclerotic preserve the shape of eye and protects it against external injury.



Front portion of sclerotic is transparent and slightly bulged outwards and known as CORNEA.

There is a layer of black tissues, below sclerotic and is called CHOROID. It serves to absorb any stray light and thus avoids blurring of the image by reflection from the eye-ball.

In front of eye, choroids merge into a coloured diaphragm known as iris which has a hole in the middle called PUPIL. When we refer to the colour of eyes of a person, we refer to the colour of iris of the person. This iris corresponds to the stop in the camera. By means of involuntary muscle control, it regulates the amount of light entering the eye. In strong light, the pupil contracts so as to admit less light whereas in dim light, it expands so as to admit as much light as required.

Behind iris is a crystalline lens with jelly-like layers. The lens is held in position with the help of *ciliary muscles*. The lens divides the eye ball into two chambers – the front chamber called anterior chamber and other between lens and the retina called posterior chamber. Anterior chamber is filled with a fluid called *aqueous humour* while the posterior chamber is filled with a jelly like substance called *vitreous humour*.

The innermost coating of the eye, covering the rear of inner surface, is a very delicate membrane called the retina. It is richly supplied with blood vessels and nerve fibres and serves as a light-sensitive screen to receive the image. The sensation of vision in the retina is carried to the brain by a nerve called *optic nerve*. The most sensitive part of retina is known as the yellow spot. It is a slightly raised portion with a slight depression known as the fovea centralis. The point where the optic nerve enters the eye is totally insensitive to light and is known as the blind spot.

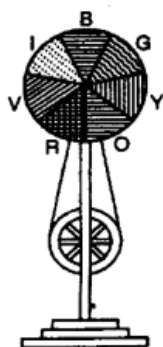
Eye Lids are provided to control the amount of light falling on the eye. Lids also protect the eyes from dust etc.

How does the eye focus: In eye, the distance between lens and the retina remains the same, while crystalline lens automatically changes its focal length by changing its curvature due to pull or push of ciliary muscles according to the distance of the object so as to bring the image to a sharp focus upon the retina. While seeing the far objects such as distant tree, the eye lens becomes thinner and flatter so as to increase its focal length. To see the objects close to the eyes, such as printed page, the lens becomes thicker so as to decrease its focal length. The process by which the eye can adapt itself to see objects at different distances is called *accommodation*.

Range of vision : The most distant point upto which a fully relaxed eye can see is called the far point. For normal eye, far point is infinity. The point at the shortest distance from the eye upto which the eye can accommodate itself and therefore, can see clearly is known as the near point. It is about 15 cm for a normal eye. The distance between far and near points is called range of vision or accommodation. Within the range of vision, there is a certain distance where the object is most clearly seen. The distance for a normal eye is about 25 cm and is known as the least distance of distinct vision.

Persistence of vision: Time for which the impression of the object seen by the eye remains on the retina even after the removal of the object is called the persistence of vision.

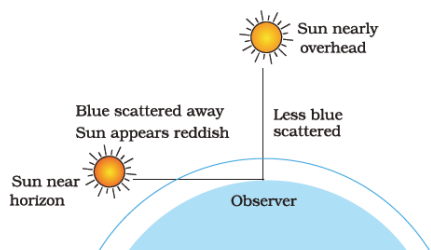
17. **Newton's colour disc:** White light consists of seven colours. This can easily be proved by Newton's colour disc. A circular disc of cardboard is painted with coloured radial bands. Order of these colours and their relative widths are the same as in the spectrum of sunlight. On rotating disc rapidly about its centre, all the colours blend and a sensation of almost white light is produced on the retina due to persistence of vision.



Since we have used pigment colours instead of spectrum colours, the disc will not appear pure white but will give a dull appearance. Pigment colours are not pure colours and hence due to impurities, the appearance of the disc is grayish instead of perfectly white.

18. The reddish appearance of the sun at sunrise/sunset is due to the scattering of light by the molecules of air and other fine particles in the atmosphere have size smaller than the wavelength of visible light from the sun near the horizon. It passes through thicker layers of air and a larger distance in the earth's atmosphere before reaching our eyes and most of the blue light and shorter wavelengths are scattered away by the particles. So, only red light, being of higher wavelength reaches us which gives the reddish appearance of the sun at sunrise or sunset.

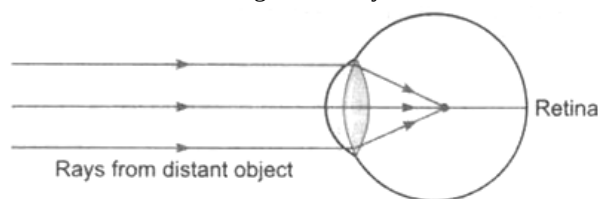
At noon, the sun appears white, not red, as only a little of the blue and violet colors are scattered as the light from the Sun overhead would travel a relatively shorter distance.



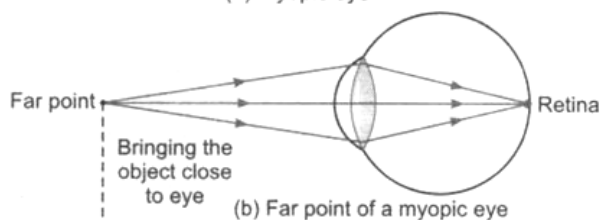
19. a. Myopia is the defect of the eye vision due to which a person can see the near by objects clearly but cannot see the far objects so distinctly.

Causes of myopia: Myopia is caused due to:

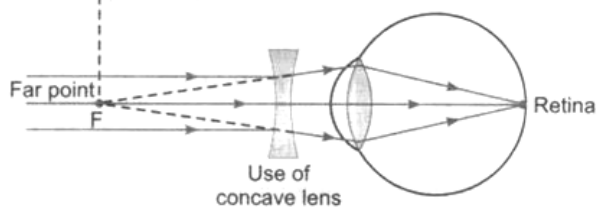
- the elongation of the eyeball.
- decrease in the focal length of the eye lens.



(a) Myopic eye



(b) Far point of a myopic eye



(c) Correction for myopia

Myopia can be corrected by using a **concave lens** of suitable focal length in the spectacles of such a person.

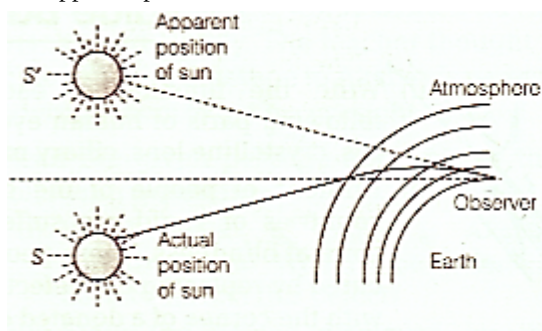
- b. The ability of the eye lens to adjust its focal length is called power of accommodation. However, the focal length of the eye lens cannot be decreased below a certain limit. The maximum accommodation of a normal eye is reached when the object is at a distance of 25 cm from the eyes. Thus, the normal eye is unable to focus an object placed within 10 cm from the eye because the nearest distance of distinct vision for a normal eye is 25cm.

20.	Dispersion of Light	Scattering of Light
	The phenomenon of splitting white light into its constituent colours by a transparent medium such as a glass prism is called dispersion of light.	The process of absorption and then reemission of light energy in different directions on interaction with the particles present in the medium is called the scattering of light.
	The dispersion of white light occurs because	Scattering of light occurs when the air molecules of size smaller than the

different colours of white light travel at different speeds while passing through the glass prism.	wavelength of incident light absorb the energy of incident light and re-emit it in different directions without change in its wavelength.
For example - The Formation of a rainbow.	For example - The blue colour of the sky.

21. The density of the atmosphere, as we know goes on decreasing as the distance above the sea level increases. Hence, the refractive index of a layer of air level goes on changing with distance. Due to this refraction takes place when light passes through the earth's atmosphere. This phenomenon is called as atmospheric refraction.

- i. Twinkling of stars: It is due to atmospheric refraction of star light. Due to variations in the refractive indices of the various layers of air, the light from a star passing through the atmosphere changes its path from time to time and therefore the amount of light reaching the eye is not always the same. This increase or decrease in the intensity of light reaching the eye results in the change in apparent position of the star. Hence, the stars appear to be twinkling.
- ii. Advanced sunrise and delayed sunset: The figure below shows the actual position of the sun S at the time of sunrise and S' the apparent position of sun. The advanced sunrise and delayed sunset is because of atmospheric refraction.



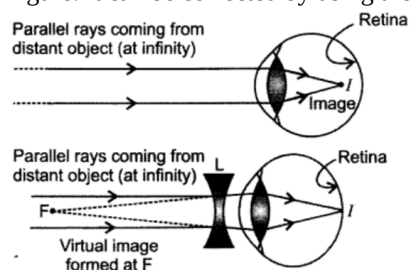
The light rays starting from the Sun travel from rarer to denser layers. They bend more and more towards the normal as it moves in denser medium.

However, an observer on earth sees an object in the direction of the rays reaching his eyes. The Sun which is actually in a position S below the horizon appears in the position S' above the horizon for him. Thus, the Sun appears to rise early by about 2 min.

Similarly, during sunset due to atmospheric refraction, the observer on earth sees the sunset than it would be without atmosphere. Thus, sunset late by about 2 min than its actual timings. These phenomena together increases the length of the day by about 4 min.

22. a. Since, 14-year old student is not able to see clearly the questions written on the blackboard placed at a distance of 5 m from him. Therefore, he is suffering from myopia (short-sightedness) defect. It is caused due to
1. excessive curvature in cornea.
 2. elongation of eyeball.

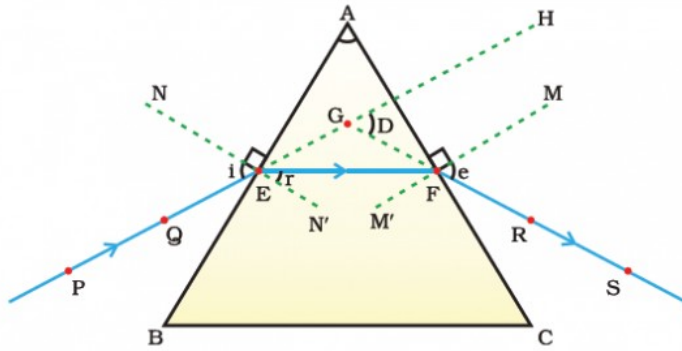
- b. A myopic eye has its far point nearer than infinity. It forms the image of a distant object in front of its retina as shown in the figure. It can be corrected by using the concave lens of suitable focal length.



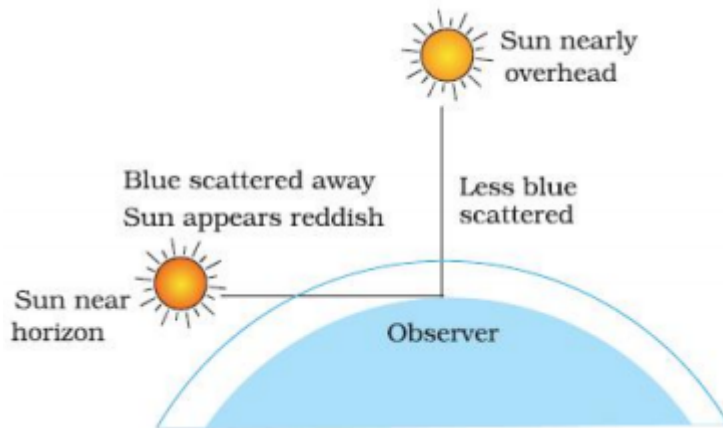
- c. The concave lens should be used to correct this defect.

23. Let PE is the incident light on the prism, EF is the refracted ray, and FS is the emergent ray. When a ray of light is entered through surface AB from air to glass, then the light ray suffers refraction and bend towards the normal ray. At the second surface AC , the rays enter from glass to air which results in bending away from the normal ray. The angle of incidence and the angle of refraction are equal. The emergent ray bend at an angle towards the direction of the incident ray. This angle is known as the Angle of deviation.

It can be shown with the following ray diagram:

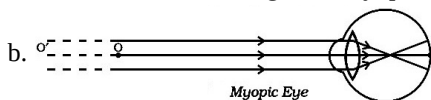


24. a. **Presbyopia:** Presbyopia is that defect of vision due to which an old person cannot see the nearby as well as the far objects clearly due to loss of power of accommodation of the eye.
Causes: The main causes of presbyopia are weakening of the ciliary muscles and diminishing flexibility of the eye lens due to old age.
Correction: It is corrected by using bi-focal lenses.
- b. In early morning, sunlight has to travel greater distance through the atmosphere to reach us. So, the shorter wavelength light scatter out. Only the longer wavelength light like red reaching us due to which sun appears red.



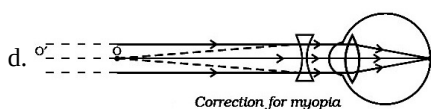
25. Light gets refracted at different layers of the atmosphere. The air just above the radiator becomes hotter than the air farther up. Since the hotter air is less dense than the cooler air above it and has less refractive index than cooler air. The physical conditions of the atmosphere are not the same, the apparent position of the object through hot air fluctuates. This is known as atmospheric refraction.
- Twinkling of stars is due to atmospheric refraction of starlight. The starlight after entering the earth's atmosphere undergoes refraction in a continuous manner before it reaches the Earth. The atmospheric refraction is due to a change in the refractive index at different levels in the atmosphere. The starlight bends towards the normal, the apparent position is different from the actual position of the star. Since the atmosphere is not stationary and keeps changing. The path of rays of light coming from star varies slightly then the apparent position of the star also varies slightly and the amount of light entering the eye flickers. Sometimes, it is brighter and sometimes the star seems fainter. In this way, stars twinkle but planets don't twinkle because planets are much closer to Earth and are seen as extended sources. If a planet is considered a collection of point sources of light, the total amount of light entering the eye on average is zero and therefore the twinkling effect of planets is nullified.

26. a. The student is suffering from myopia.

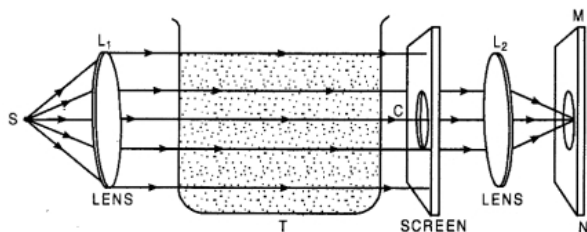


- c. Concave lens

- Decrease in the focal length of the eye lens
- Eye ball gets elongated.



27. We observe that the sky appears blue, there is a reddening of sky at sunrise and at sunset. The path of beam of light becomes visible when it passes through space containing smoke, tiny water droplets, suspended dust particles. The path of light is visible when it pass through a medium the size of whose particle are comparable to wavelength of light.



Colour of scattered light depends upon the size of scattering particles. Very small particles scatter blue light while larger particles scatter light of longer wavelength. The following experiment will make the point clear.

When a little of sulphuric acid is put in a dilute solution of sodium thiosulphate (hypo), sulphur particles begin to form. The particles grow slowly with time.

Experiment: Take a white source of light (S) at focus of converging lens L_1 . The beam is rendered parallel. Allow it to pass through tank T containing pure water. Allow the beam to pass through a hole C and then converge to screen MN by a converging lens L_2 . A sharp bright image will be formed on MN.

Dissolve about 100 g of hypo in a litre of pure water. Put it in tank T and add about 1 ml (1 cm^3) of concentrated sulphuric acid in it. You will observe blue light from three sides of the glass tank. This is due to scattering of smaller wavelengths of light by minute colloidal particles of sulphur formed. However, in fourth side facing the circular hole C will first appear orange-red colour and then bright crimson red colour on the screen.

28. The defect caused due to gradual weakening of ciliary muscles and diminishing flexibility of the eye lens is presbyopia.

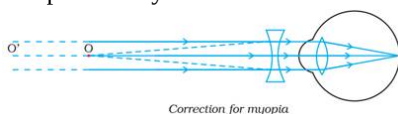
Presbyopia is the defect of eye in which a person cannot see nearby objects comfortably and distinctly without corrective eye glasses. A presbyopic eye has its near point greater than 25 cm and is gradually increases as the eye becomes older. The type of lens required by such person to improve the vision is bifocal lens.

A bifocal lens consists of both convex lens and concave lenses. In this bifocal lens the convex lens is used to correct hypermetropia (far sightedness) to see distinct object clearly and concave lens is used to correct myopia (short sightedness) to see nearby object clearly.

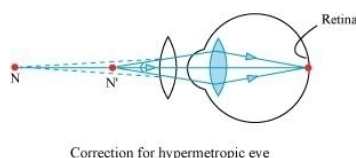
29. When a person is unable to clearly see distant objects, he is considered a myopic person. Such a person is suffering from myopia or short-sightedness or nearsightedness. This happens when image is formed in front of the retina.

When a person is unable to clearly see a nearby object, he is considered a hypermetropic person. Such a person is suffering from hypermetropia or longsightedness. This happen when image is formed behind the retina but not on it.

Correction of Myopia: A person suffering from myopia needs to use a concave lens of suitable focal length. The concave lens diverges the rays coming from infinity. After refraction from the concave lens, the rays appear to be coming from the far point of this person's eye.



Correction of Hypermetropia: A hypermetropic person needs to use a convex lens of suitable focal length. The convex lens converges the light rays coming from a nearby object. As a result, these light rays appear to be coming from the near point of this person's eyes. Due to this, a clear image of nearby object is made on the retina of that person.



30. (i) Power, $P = -5.5 \text{ D}$, Focal length, $f = ?$

$$P = \frac{1}{f(\text{in m})}$$

$$\text{or } -5.5 = \frac{1}{f}$$

$$\text{or } f = \frac{1}{-5.5}$$

$$= -0.18 \text{ m for correcting the ditant vision.}$$

The negative (-) sign indicates that the lens is concave.

(ii) Power, $P = +1.5 \text{ D}$

Focal length, $f = ?$

$$P = \frac{1}{f(\text{in m})}$$

$$\text{or } f = \frac{1}{+1.5}$$

or $f = +0.67 \text{ m}$ for correcting the near vision.

The positive sign indicates that the lens is convex.

31. a. Defect of vision - Myopia or short sightedness or near-sightedness.

Causes of myopia:

- i. Excessive curvature of eye lens.
- ii. Elongation of eye ball

Methods of correction: By using a concave lens of suitable power.

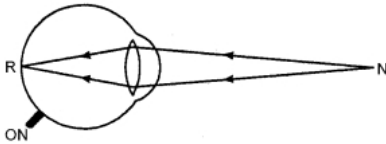
b. Due to atmospheric refraction:- Due to atmospheric refraction the star light reaching our eyes increases and decreases continuously. And the star appears to Twinkle at night.

32. The person is suffering from hypermetropia or long-sightedness.

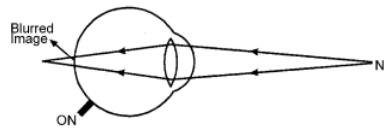
A long sighted person can see distant objects clearly, but cannot see distinctly objects lying closer than a certain distance. Thus, he cannot see clearly an object lying at 25 cm which is the least distance of distinct vision for a normal eye. His far point is farther from the eye than 25 cm, say at 75 cm or so.

Causes of the defect: This defect is due to either :

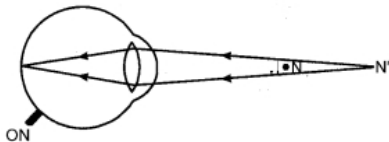
- i. the size of the eye ball becoming too short.
- ii. the lens becoming too thin, so that its focal length becomes abnormally large.



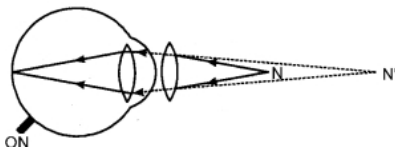
Normal eye : It is able to focus the rays from normal near point (25 cm from eyes)



Defective eye: Due to eye ball getting short or an increase in focal length of eye lens, the rays do not focus on retina.



Defective eye: is able to form image at retina when object is moved from N to N' the near point of defective eye.



Corrected eye : A convex (or any convergent) lens of suitable focal length converges the rays to match those coming from N'.

Hypermetropic eye.

Normal eye is able to focus on retina the rays emerging out from N.

However, the defective eye is not able to focus the rays from near point of normal eye i.e. N.

It can focus the rays from near point of defective eye. i.e. N'.

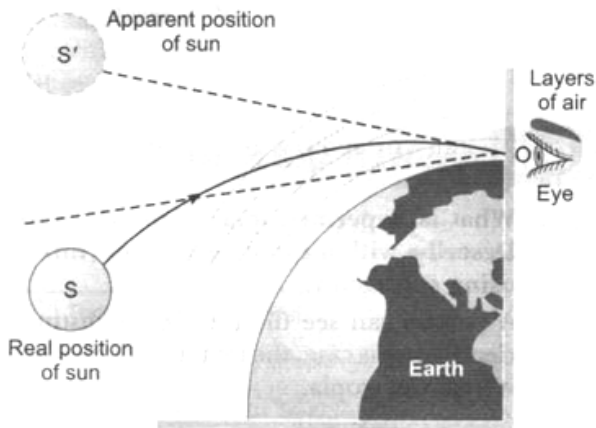
From fig. we conclude that more inclined rays are not focused on retina whereas less inclined rays from N' get focused on retina.

Correction of this Defect: Such a defect is corrected by placing a convex or converging lens of suitable focal length before the eye so that the rays diverging from N appear to come, after refraction, from the near point N'.

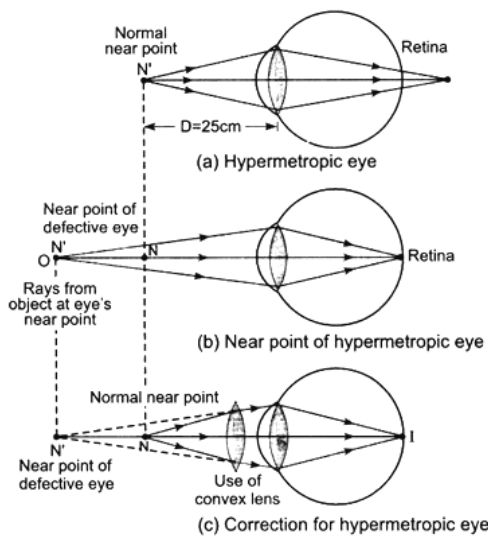
33. The layers of air nearer to earth are denser than those above it. At sunrise and sunset when the sun is below the horizon, the light rays starting from sun are falls on these layers. They pass through successively denser layers and thus get bent more and more towards the normal until they fall upon the eye of the observer O.

To the observer O, these rays appear to come from S' which is above horizon. It is for this reason that the sun is visible to us a little before it rises above the horizon and so also can be seen for more time as it sets below the horizon. The difference of time is

about two minutes each for early rise and late setting of the Sun.



34. Hypermetropia refers to the condition of eyesight defect, when a person can see distant objects distinctly but cannot see nearby objects so clearly. This is also termed as long sightedness, it occurs when light from near objects is not quite brought to focus in time to hit the retina. The point of focus would in fact be behind the retina.

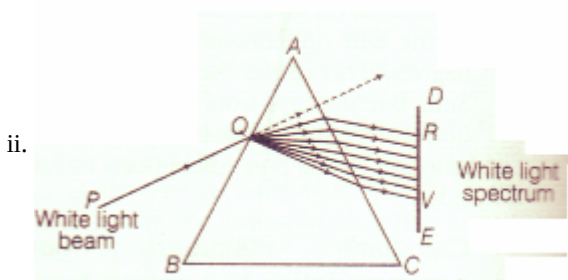


Causes of hypermetropia: This defect arises because of either,

- i. the focal length of the eye lens is too long, or
- ii. the eyeball becomes too short, so that light rays from the nearby object, say at point N, cannot be brought to focus on the retina to give a distinct image.

Hypermetropia can be corrected by using convex lens of suitable focal length in spectacles.

35. i. The phenomenon of splitting of a beam of white light into its seven constituent colours when passed through a transparent medium is known as dispersion, which was first discovered by Issac Newton in 1666. To understand this phenomenon, let us take a thick sheet of cardboard and make a small hole or narrow slit in its middle allow sunlight to fall on the slit. This gives a narrow beam of white light. Now, take a glass prism and allow this white light to fall on one of its faces as shown in figure. Turn the prism slowly until the light that comes out of it appears on a nearby screen. We see a beautiful band of seven colours on the screen called visible spectrum. The sequence of colours seen from the lower part of the screen is violet (V), indigo (I), blue (B), green (G), yellow (Y), orange (O) and red (R). The acronym for this is VIBGYOR.

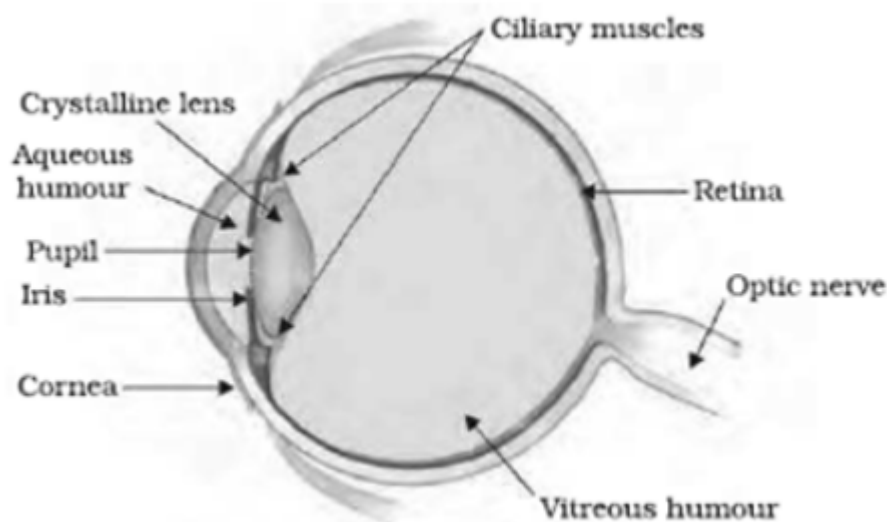


- ii. Red light has the maximum wavelength and violet light has the minimum

wavelength, so in a transparent medium except air and vacuum, red light having the largest wavelength suffers the least deviation while violet light having the least wavelength bends the most.

$$\text{As wavelength} \propto \frac{1}{\text{deviation (or bending)}}$$

36. The structure of the human eye is as follows:



The function of the human eye:

The human eye acts as a camera. The lens system of eye forms image on Retina which is a light-sensitive screen. Light is entered through a thin membrane known as the cornea. It forms the transparent bulge on the front surface of the eyeball. The eyeball is of diameter 2.3 cm. The maximum refraction of light is refracted at the outer surface of the cornea of the eye. The crystalline lens provides adjustment of focal length which is required to focus different images on the retina. Iris is a dark muscular diaphragm that controls the size of the pupil. Light entering the eye is controlled by the iris. An inverted, real image is formed by the lens on the retina. The retina is a delicate membrane that consists of a number of light-sensitive cells. Upon illumination, the light-sensitive cells get activated and electrical signals are generated. These signals are then sent to the brain through optic nerves. The signals are interpreted by the brain and the information is processed and we can see the objects.

37.

Presbyopia	Cataract
It is a defect of vision that occurs in old age due to which a person cannot see the nearby objects clearly due to loss of power of accommodation of eye.	It is a medical condition in which the lens of the eye of a person becomes progressively cloudy resulting in blurred vision.
It can be corrected by using spectacles having a bifocal lens of suitable power.	It cannot be corrected by any type of spectacle lenses.
Surgical operations are not required to rectify presbyopia.	The vision of a person can be restored by getting surgical operation.